SIMATS ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Jotheesh A
192511029RD

QUESTIONS

Q1
Sum & Average - Tuple
10 marks

Q2
Missing Number - Tuple
10 marks

Q3
Maximum Element - Tuple
10 marks

Q4
Remove Duplicate Elements - Tuple
10 marks

Q5
Peak Element - List
10 marks

Q6
Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q1 **Sum & Average - Tuple**

10 marks

Write a Python program to find the sum and average of the elements in a tuple.

INPUT:

10 20 30 40 50

OUTPUT:

150

30.0

Your Code

```
values = tuple(map(int, input().split()))

total = sum(values)
average = total / len(values)

print(total)
print(average)
```

Input

```
10 20 30 40 50
```

Output

```
150
30.0
```

Visible Test Cases

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Q2
Missing Number - Tuple
10 marks

Q3
Maximum Element - Tuple
10 marks

Q4
Remove Duplicate Elements - Tuple
10 marks

Q5
Peak Element - List
10 marks

Q6
Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q8
Majority Element -List
10 marks

1 2 3 5 6
OUTPUT:
4

Your Code

```python
nums = tuple(map(int, input().split()))
n = len(nums) + 1
total = n * (n + 1) // 2
missing = total - sum(nums)
print(missing)
```

Input

```
1 2 3 5 6
```

Output

```
4
```

Visible Test Cases

Test Case 1

10.00 / 10 marks

Submitted

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Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Write a Python program to find the largest element in a tuple.

INPUT:

12 45 8 67 23

OUTPUT:

67

Your Code

```
values = tuple(map(int, input().split()))
print(max(values))
```

Input

```
12 45 8 67 23
```

Output

```
67
```

Visible Test Cases

Test Case 1

10.00 / 10 marks

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Peak Element - List
10 marks

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Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q8
Majority Element -List
10 marks

Q9

OUTPUT:
(40, 10, 50, 20, 30)

Your Code

```
values = tuple(map(int, input().split()))

result = tuple(set(values))

print(result)
```

Input

```
10 20 30 20 40 10 50
```

Output

```
(40, 10, 50, 20, 30)
```

Visible Test Cases

Test Case 1

10.00 / 10 marks

Submitted

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Peak Element - List
10 marks

Q6
Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q8
Majority Element -List
10 marks

Q9

1 3 20 4 1 0

OUTPUT:

20

Your Code

```
def find_peak(arr):
    n = len(arr)
    for i in range(n):
        left_ok = (i == 0) or (arr[i] >= arr[i - 1])
        right_ok = (i == n - 1) or (arr[i] >= arr[i + 1])
        if left_ok and right_ok:
            return arr[i]
    return -1

nums = list(map(int, input().split()))
print(find_peak(nums))
```

Input

```
1 3 20 4 1 0
```

Output

```
20
```

Visible Test Cases

Test Case 1

10.00 / 10 marks

Submitted

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Maximum Element - Tuple
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Remove Duplicate Elements - Tuple
10 marks

Q5
Peak Element - List
10 marks

Q6
Second Largest and Second Smallest - Tuple
10 marks

Q7

Q6 **Second Largest and Second Smallest - Tuple** 10 marks

Write a Python program to find the second largest and second smallest elements in a list using a tuple.

INPUT:

10 20 5 40 15 30

OUTPUT:

10

30

Your Code

```
nums = list(map(int, input().split()))

unique_sorted = sorted(set(nums))

second_smallest = unique_sorted[1]
second_largest = unique_sorted[-2]

print((second_smallest, second_largest)[0])
print((second_smallest, second_largest)[1])
```

Input

```
10 20 5 40 15 30
```

Output

```
10
30
```

Visible Test Cases

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Peak Element - List
10 marks

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Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q8
Majority Element -List
10 marks

Q9
Lucky Number

Q7 Element Exists -Tuple

10 marks

Write a Python Program to Check Whether an Element Exists in a Tuple
INPUT1:
10 20 30 40 50
30
Output 1
Element found
INPUT2:
10 20 30 40 50
25
Output 1
Element not found

Your Code

```
arr = tuple(map(int, input().split()))

x = int(input())

if x in arr:
    print("Element found")
else:
    print("Element not found")
```

Input

```
10 20 30 40 50
30
```

Output

```
Element found
```

Visible Test Cases

Test Case 1

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Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q8
Majority Element -List
10 marks

Q9
Lucky Number

Q8 Majority Element -List

10 marks

Write a Python function to check whether a list contains a majority element.
INPUT:
2 2 1 2 3 2 2
OUTPUT:
2

Your Code

```python
def majority_element(arr):

    candidate = None
    count = 0

    for x in arr:
        if count == 0:
            candidate = x
            count = 1
        elif x == candidate:
            count += 1
        else:
            count -= 1

    if arr.count(candidate) > len(arr) // 2:
        return candidate
    return None

arr = [2, 2, 1, 2, 3, 2, 2]
print(majority_element(arr))
```

Input

2 2 1 2 3 2 2

Output

2

Visible Test Cases

Test Case 1

10.00 / 10 marks

Submitted

Q2
Missing Number - Tuple
10 marks

Q3
Maximum Element - Tuple
10 marks

Q4
Remove Duplicate Elements - Tuple
10 marks

Q5
Peak Element - List
10 marks

Q6
Second Largest and Second Smallest - Tuple
10 marks

Q7
Element Exists -Tuple
10 marks

Q8
Majority Element -List
10 marks

Q9
Lucky Number
10 marks

Q10
Frequency of Each Element - List
10 marks

34
OUTPUT1:
Lucky Number
INPUT2:
123
OUTPUT2:
Not a Lucky Number

Your Code

```python
def is_lucky(n):

    if n < 10:
        return n == 7
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    return is_lucky(s)

num = int(input().strip())

if is_lucky(num):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

Input

123

Output

```
Not a Lucky Number
```

Visible Test Cases

Test Case 1

Test Case 2

10.00 / 10 marks

Submitted

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QUESTIONS

- Q1 Sum & Average - Tuple 10 marks
- Q2 Missing Number - Tuple 10 marks
- Q3 Maximum Element - Tuple 10 marks
- Q4 Remove Duplicate Elements - Tuple 10 marks
- Q5 Peak Element - List 10 marks
- Q6 Second Largest and Second Smallest - Tuple 10 marks
- Q7 Element Exists -Tuple 10 marks
- Q8 Majority Element -List 10 marks
- Q9 Lucky Number

Q10 **Frequency of Each Element - List** 10 marks

Write a Python program to find the frequency of each element in an array using List.

INPUT:

1 2 3 2 4 1 5 2

OUTPUT:

1 = 2

2 = 3

3 = 1

4 = 1

5 = 1

Your Code

```
arr = list(map(int, input().split()))

freq = {}
for x in arr:
    freq[x] = freq.get(x, 0) + 1

for num in sorted(freq):
    print(f"{num} = {freq[num]}")
```

Input

```
1 2 3 2 4 1 5 2
```

Output

```
1 = 2
2 = 3
3 = 1
4 = 1
5 = 1
```

Visible Test Cases

Test Case 1

10.00 / 10 marks

Submitted